

EX. NO : 09

PERSISTENCE LAYER (ROOM DATABASE)

AIM

To understand the Persistence Layer in Android application architecture and implement data storage using Room Database.

PROCEDURE

1. Open **Android Studio** and create a new project → Empty Activity → Kotlin.
2. Add Room dependencies in **build.gradle** and sync project.
3. Design UI using EditText, Button, and TextView.
4. Create **Entity class (UserEntity)**.
5. Create **DAO interface (UserDao)**.
6. Create **Database class (AppDatabase)**.
7. Write logic in **MainActivity.kt** to insert and retrieve data.
8. Run the application and verify stored data persists.

CODE

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <EditText
        android:id="@+id/etName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Name"/>
```

```

<Button
    android:id="@+id/btnSave"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Save to Database"/>
<TextView
    android:id="@+id/tvResult"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Stored Data:"
    android:textSize="18sp"
    android:paddingTop="20dp"/>
</LinearLayout>

```

UserEntity.kt

```

package com.example.persistencedemo
import androidx.room.Entity
import androidx.room.PrimaryKey
@Entity(tableName = "user_table")
data class UserEntity(
    @PrimaryKey(autoGenerate = true)
    val id: Int = 0,
    val name: String
)

```

UserDao.kt

```

package com.example.persistencedemo
import androidx.room.Dao
import androidx.room.Insert
import androidx.room.Query
@Dao
interface UserDao {
    @Insert
    suspend fun insertUser(user: UserEntity)
}

```

```

    @Query("SELECT * FROM user_table")
    suspend fun getAllUsers(): List<UserEntity>
}

```

AppDatabase.kt

```

package com.example.persistencedemo

import android.content.Context
import androidx.room.Database
import androidx.room.Room
import androidx.room.RoomDatabase

@Database(entities = [UserEntity::class], version = 1)
abstract class AppDatabase : RoomDatabase() {
    abstract fun userDao(): UserDao

    companion object {
        private var INSTANCE: AppDatabase? = null

        fun getDatabase(context: Context): AppDatabase {
            return INSTANCE ?: synchronized(this) {
                Room.databaseBuilder(
                    context.applicationContext,
                    AppDatabase::class.java,
                    "user_database"
                ).build().also {
                    INSTANCE = it
                }
            }
        }
    }
}

```

MainActivity.kt

```

package com.example.persistencedemo

import android.os.Bundle
import android.widget.Button
import android.widget.EditText

```

```

import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity
import kotlinx.coroutines.CoroutineScope
import kotlinx.coroutines.Dispatchers
import kotlinx.coroutines.launch
class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)
        val etName = findViewById<EditText>(R.id.etName)
        val btnSave = findViewById<Button>(R.id.btnSave)
        val tvResult = findViewById<TextView>(R.id.tvResult)
        val db = AppDatabase.getDatabase(this)
        btnSave.setOnClickListener {
            val name = etName.text.toString()
            CoroutineScope(Dispatchers.IO).launch {
                db.userDao().insertUser(UserEntity(name = name))
                val users = db.userDao().getAllUsers()
                runOnUiThread {
                    tvResult.text = "Stored Data:\n"
                    users.forEach {
                        tvResult.append(it.name + "\n")
                    }
                }
            }
        }
    }
}

```

OUTPUT

Persistence Demo

Priyanga

Save

Stored Data:
Priyanga

RESULT

Thus, the Persistence Layer was successfully implemented using Room Database, enabling storage and retrieval of user data in an Android application.